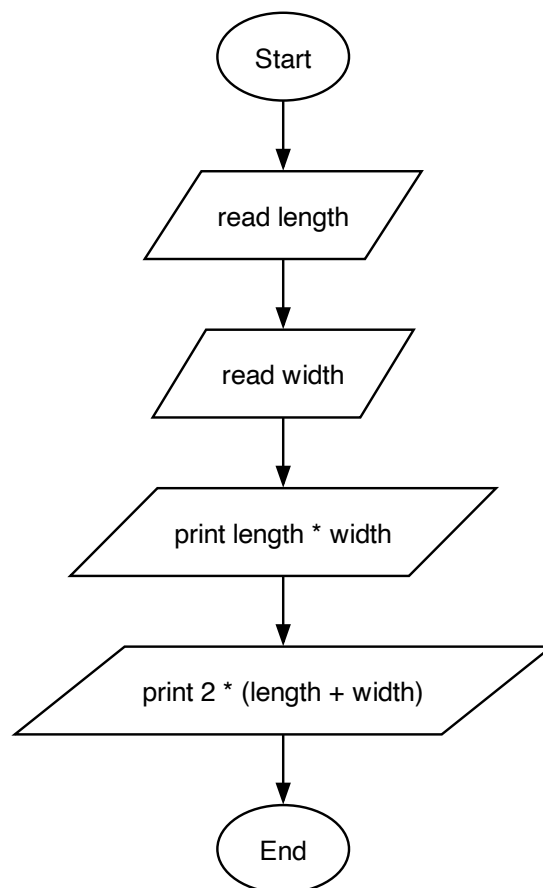


*Answer key*

**Problem 1.**

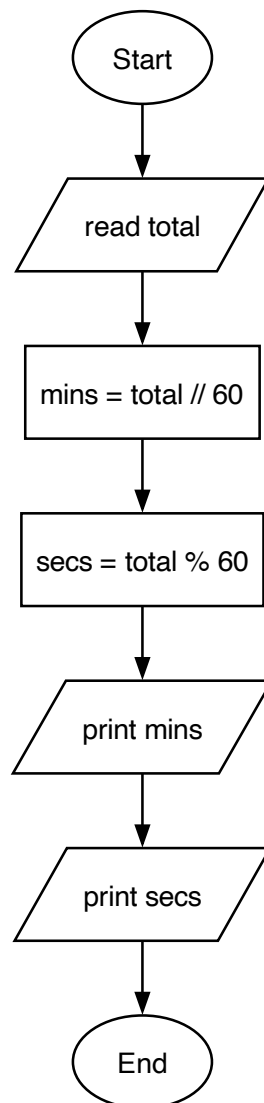


## Problem 2.

(i) Expected output for each input:

| input <i>total</i> | output |
|--------------------|--------|
| 200                | 3, 20  |
| 95                 | 1, 35  |
| 60                 | 1, 0   |

(ii) Flowchart:

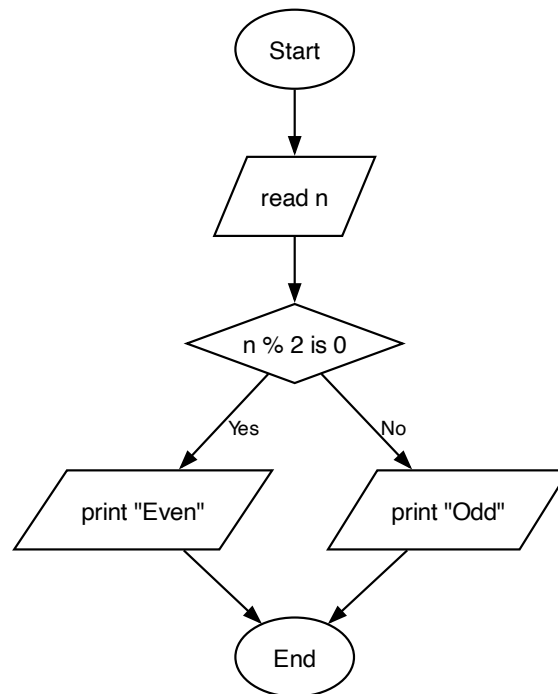


### Problem 3.

(i) Expected output for each input:

| input $n$ | output |
|-----------|--------|
| 7         | Odd    |
| 10        | Even   |
| 0         | Even   |

(ii) Flowchart:

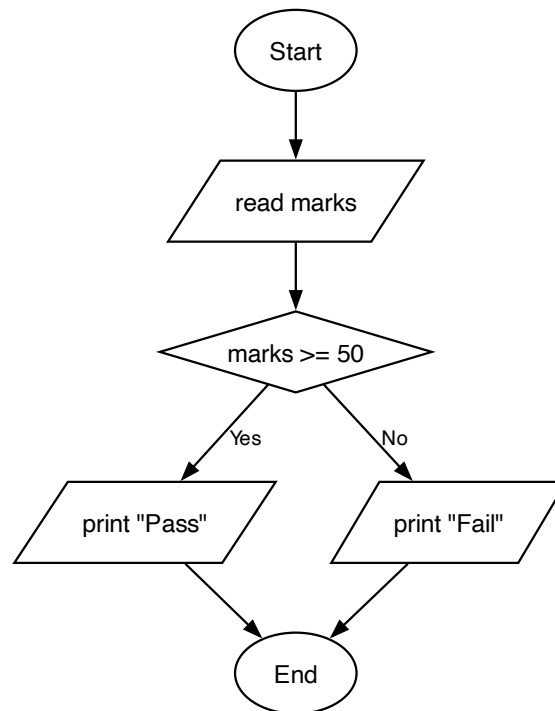


#### Problem 4.

(i) Expected output for each input:

| input <i>marks</i> | output |
|--------------------|--------|
| 72                 | Pass   |
| 50                 | Pass   |
| 38                 | Fail   |

(ii) Flowchart:

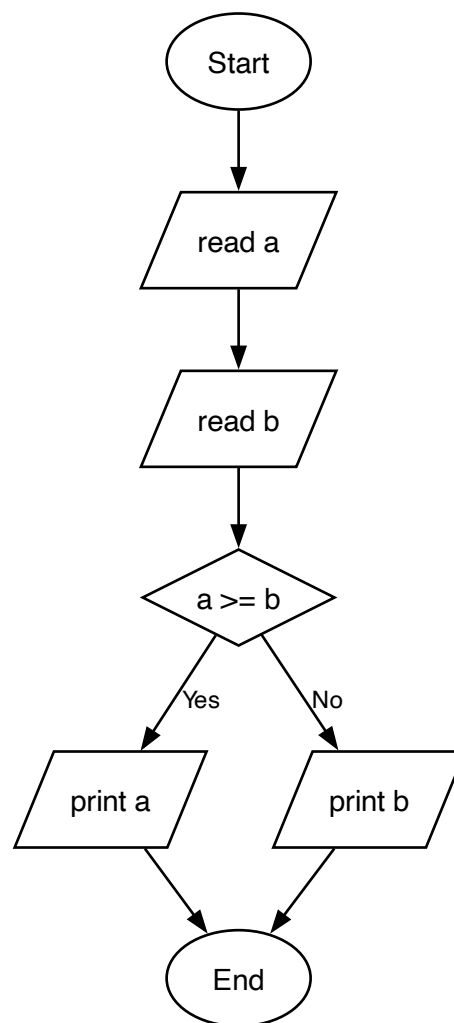


### Problem 5.

(i) Expected output for each input:

| input $a$ | input $b$ | output |
|-----------|-----------|--------|
| 8         | 3         | 8      |
| 4         | 9         | 9      |
| 6         | 6         | 6      |

(ii) Flowchart:

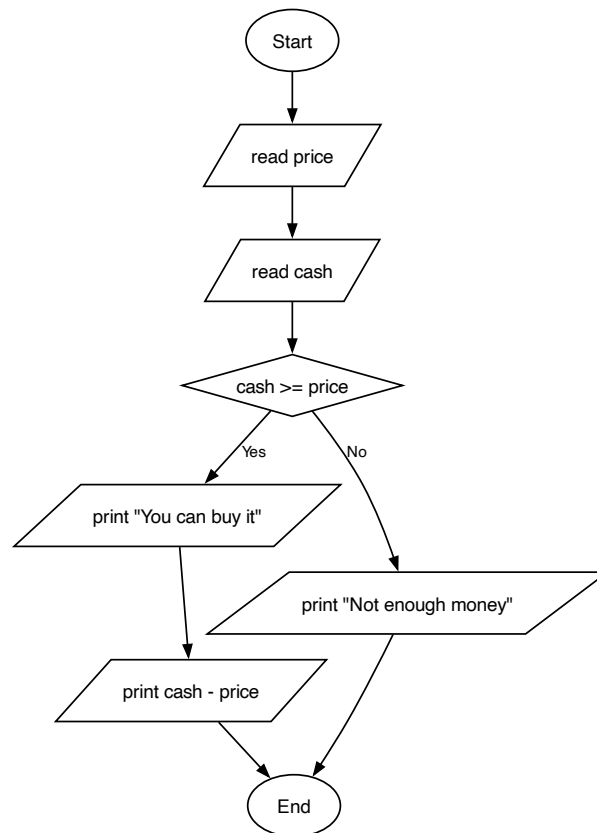


## Problem 6.

(i) Expected output for each input:

| input <i>price</i> | input <i>cash</i> | output             |
|--------------------|-------------------|--------------------|
| 50                 | 80                | You can buy it, 30 |
| 50                 | 50                | You can buy it, 0  |
| 50                 | 20                | Not enough money   |

(ii) Flowchart:

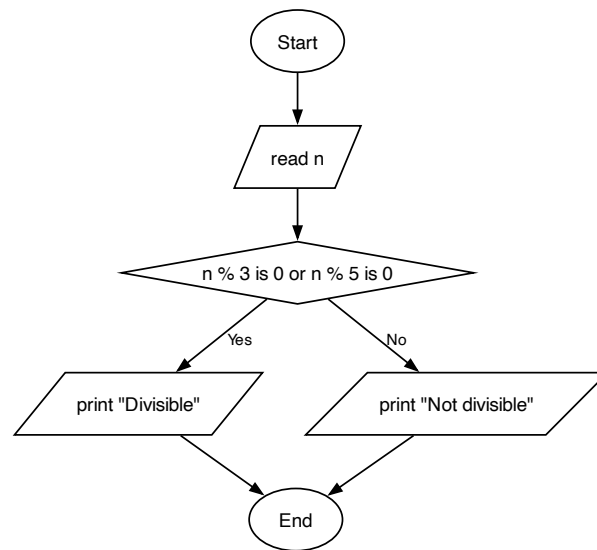


### Problem 7.

(i) Expected output for each input:

| input $n$ | output        |
|-----------|---------------|
| 9         | Divisible     |
| 20        | Divisible     |
| 7         | Not divisible |

(ii) Flowchart:

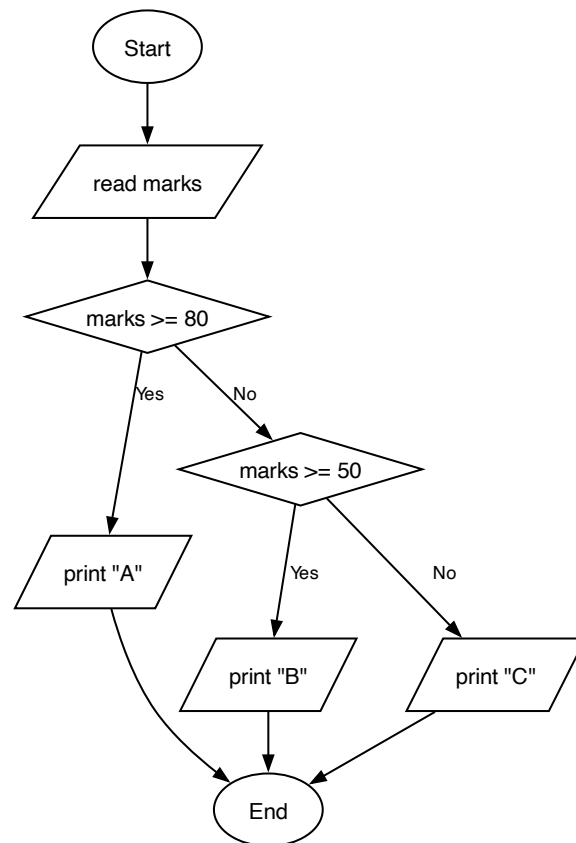


### Problem 8.

(i) Expected output for each input:

| input <i>marks</i> | output |
|--------------------|--------|
| 91                 | A      |
| 64                 | B      |
| 42                 | C      |

(ii) Flowchart:

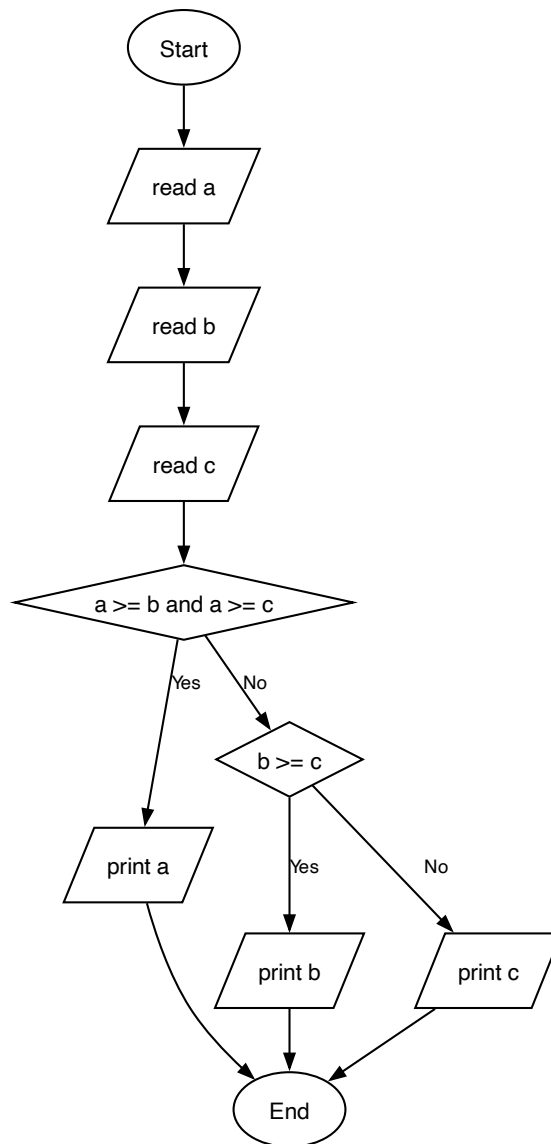


### Problem 9.

(i) Expected output for each input:

| input $a$ | input $b$ | input $c$ | output |
|-----------|-----------|-----------|--------|
| 4         | 9         | 2         | 9      |
| 7         | 1         | 5         | 7      |
| 3         | 3         | 8         | 8      |

(ii) Flowchart:

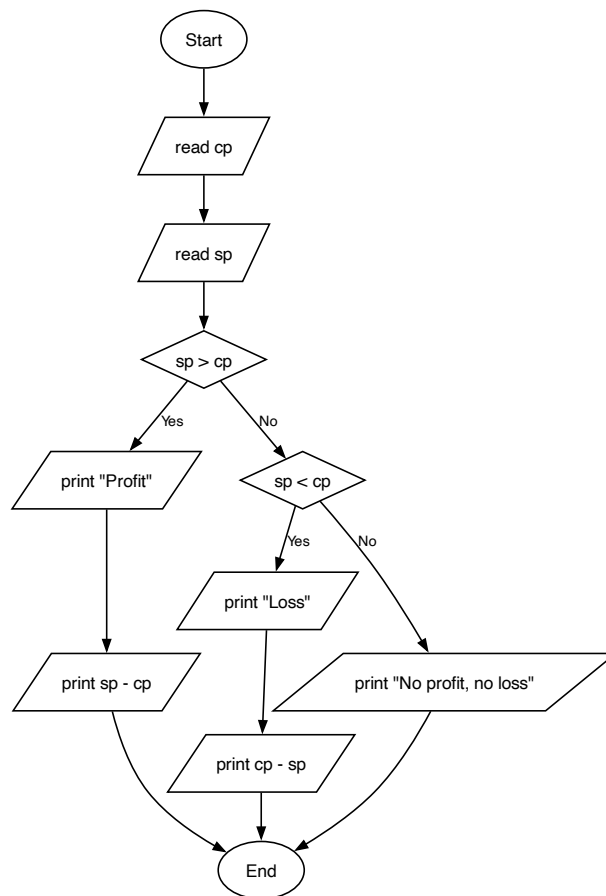


### Problem 10.

(i) Expected output for each input:

| input $cp$ | input $sp$ | output             |
|------------|------------|--------------------|
| 100        | 130        | Profit, 30         |
| 100        | 80         | Loss, 20           |
| 100        | 100        | No profit, no loss |

(ii) Flowchart:

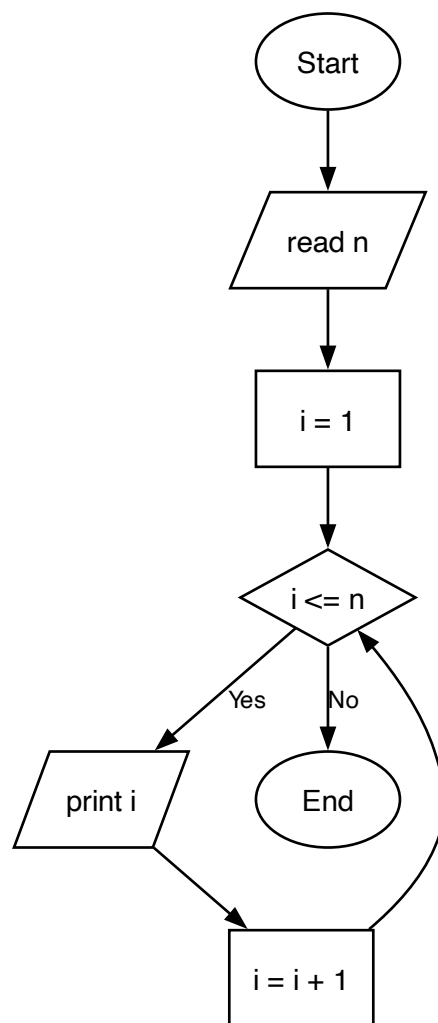


**Problem 11.**

(i) Expected output for each input:

| input $n$ | output        |
|-----------|---------------|
| 5         | 1, 2, 3, 4, 5 |
| 1         | 1             |
| 3         | 1, 2, 3       |

(ii) Flowchart:

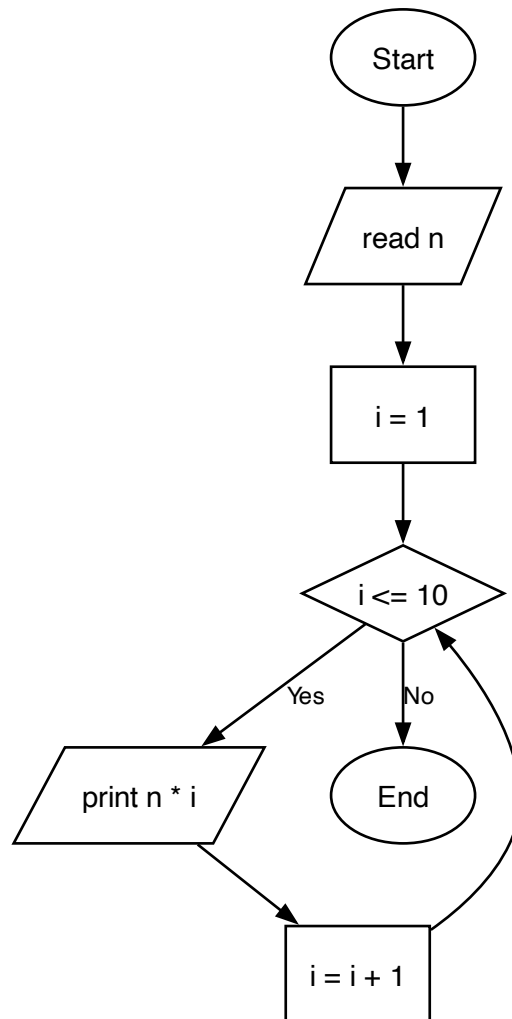


**Problem 12.**

(i) Expected output for each input:

| input $n$ | output                                |
|-----------|---------------------------------------|
| 7         | 7, 14, 21, 28, 35, 42, 49, 56, 63, 70 |
| 3         | 3, 6, 9, 12, 15, 18, 21, 24, 27, 30   |

(ii) Flowchart:

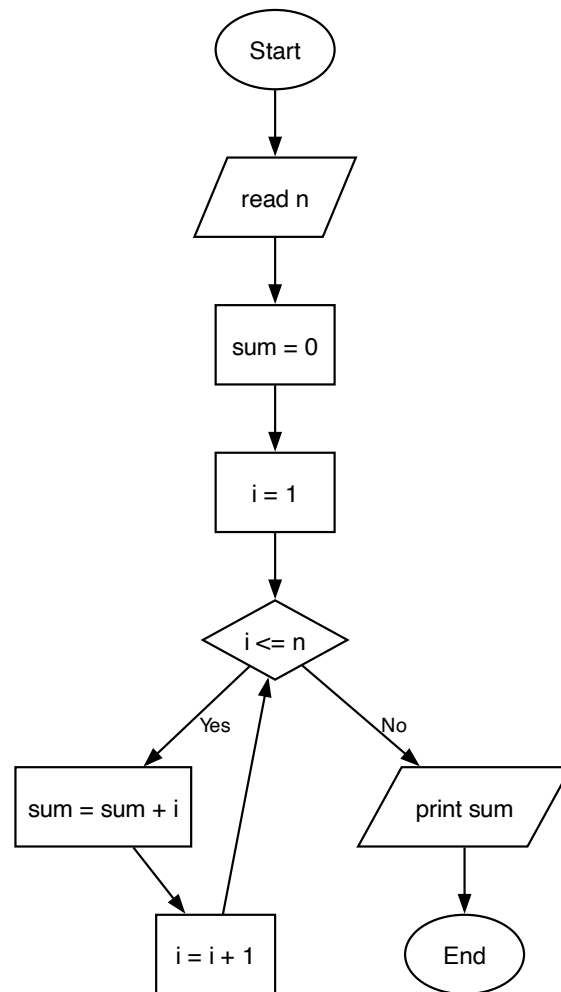


**Problem 13.**

(i) Expected output for each input:

| input $n$ | output |
|-----------|--------|
| 4         | 10     |
| 1         | 1      |
| 6         | 21     |

(ii) Flowchart:

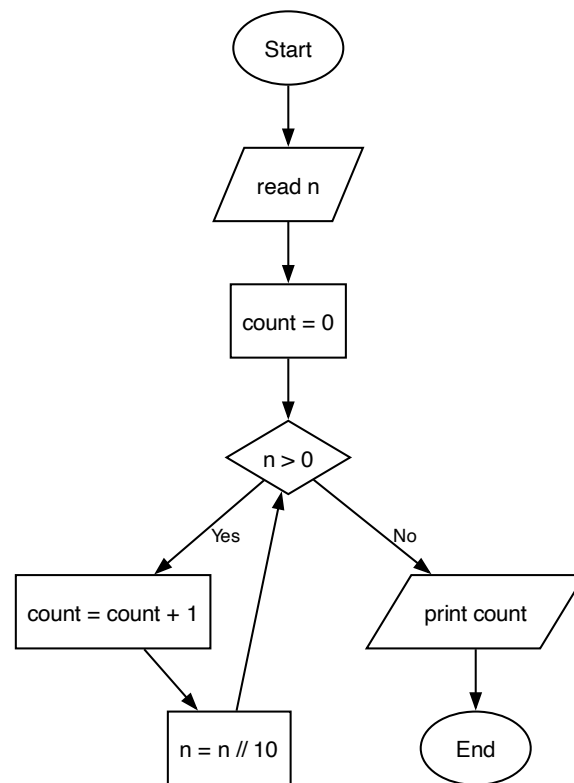


### Problem 14.

(i) Expected output for each input:

| input $n$ | output |
|-----------|--------|
| 4072      | 4      |
| 538       | 3      |
| 9         | 1      |

(ii) Flowchart:



**Problem 15.**

(i) Expected output for each input:

| input $n$ | output     |
|-----------|------------|
| 8         | 2, 4, 6, 8 |
| 5         | 2, 4       |
| 1         |            |

(ii) Flowchart:

